FACULTY OF ENGINEERING OF THE UNIVERSITY OF PORTO

Systems Engineering

Team Meeting Report

SWAID — Plate Resonance (PR)

Meeting Date:	March 12, 2026
Location:	FEUP I-108
Time:	16:10 — 17:45
Proposed by:	Professor Sérgio Reis Cunha

1 Attendance & Logistics

Team Member	Present	Role / Focus
Miguel Amorim	[P]	Team Leader
João Oliveira	[P]	Technical Leader (HW)
Artur Kovbasa	[P]	Technical Leader (SW)
Afonso Cruz	[P]	Member
Alexandre Gorra	[P]	Member
Beatriz Martins	[F]	Member
Hugo Gomes	[P]	Member
João Pinto	[P]	Member
Lucas Aparicio	[P]	Member

2 Meeting Objectives & Discussion Notes

2.1 Chladni Symbol Alphabet Methodology

Discussion & Notes:

- Evaluate the simulator's accuracy by cross-referencing results with established physical benchmarks.
- Transition from wave-pattern rendering to discrete particle visualization to improve symbol identification.
- Utilize complex plate geometries to increase modal density and expand the variety of achievable patterns.
- Implement independent phase control for each transducer to enable asymmetric pattern generation.
- Develop automated simulation sweeps across multiple configurations to catalog a standardized symbol alphabet.
- Analyze the influence of frequency, plate geometry, transducer phase, and placement symmetry on resonant mode stability.

Conclusions:

- Verify physical formulations to ensure generated images accurately correspond to the input frequencies.
- Automate parameter sweeps for transducer positioning, frequency, and phase to identify optimal resonance modes.
- Standardize material selection and identify commercially available dimensions for the final plate design.
- Investigate the relationship between damping factors and the total number of achievable modes.

- Incorporate a border-height parameter into the simulation environment to better represent physical constraints.
- Implement a particle “re-centering” mechanism and explore performance optimization via C/C++ porting.
- Finalize plate specifications (thickness, dimensions, material) and transducer count/positioning by the next milestone.
- Identify specific symbols that intuitively represent interactive hand gestures (e.g., directional shifting of sand).

2.2 Illumination Effects

Discussion & Notes:

- Proposal 1: Infinity Mirror effect for visual depth.
- Proposal 2: Fluorescent Powder paired with UV (Black-Light) illumination.
- Proposal 3: Integrated LED Matrix for dynamic background lighting.
- Proposal 4: Side-mounted LED strips for high-contrast pattern highlights.

Conclusions:

- The Infinity Mirror concept was approved; coordination with the PR team for implementation is required.
- Define specifications for LED Matrices, LED Strips, UV lighting, and Fluorescent powder to finalize the procurement budget.

2.3 FPGA Platform Selection & Software Environment

Discussion & Notes:

- Analyze sound card capabilities and hardware-software interface requirements.
- Research ASIO driver integration for low-latency audio processing.
- Investigate communication protocols between the Raspberry Pi and external sound/FPGA hardware.
- Perform a comparative analysis between professional sound card solutions and custom FPGA implementations regarding cost and flexibility.

Conclusions:

- Procurement of the **Sipeed Tang Nano 9K** FPGA platform.
- Draft a technical specification document for the Tang Nano 9K covering:

- PLL channel availability, LUT resources, and amplitude control strategies.
- Implementation logic for Phase Accumulators and Sigma-Delta modulators.
- Integration scheme for the RPi-to-FPGA connection and maximum clock signal thresholds.
- Evaluation of sound card control features and their reproducibility within the FPGA logic.

2.4 Sound Card Implementation

Discussion & Notes:

- Evaluate the feasibility of using ASIO drivers for high-resolution waveform control.

Conclusions:

- Acquire high-quality audio jacks and a sound card with compatible RPi connectivity.
- Test waveform resolution and latency using ASIO drivers.
- Attempt dual-waveform generation from a single output to assess parameter granularity.

2.5 Weekly Deliverable: Market Survey

Discussion & Notes:

- Market segmentation focusing on existing acoustic visualization and resonance systems.
- Identification of relevant state-of-the-art technologies and evaluation criteria.
- Sourcing and verification of industrial and academic data.
- Definition of the data collection methodology and comparative analysis structure.

Conclusions:

- João Pinto and Beatriz Martins are assigned to draft the Market Survey document.
- Review ESIS reference materials and high-scoring submissions from previous years.

2.6 Presentation

Discussion & Notes:

- Design the theoretical framework for the team's upcoming classroom topic presentation.

Conclusions:

- Miguel, Lucas, and João Thomaz are assigned to the presentation

3 Task Assignment

*Categories: Management (**G**), Deliverables (**E**), Development (**D**).*

Task Description	Cat.	Assigned To	Status	Deadline
Budget Elaboration & Hardware Selection	D	Miguel	Open	13/03
Sound Card Performance Evaluation	D	Afonso	Open	18/03
FPGA Architectural System Design	G	Miguel, Lucas, Afonso, João O.	Open	18/03
Finalize Chladni Simulation Environment	D	Artur, Alexandre, Hugo	Open	11/03
Market Survey Document (Deliverable #3)	E	João Pinto, Beatriz	Open	13/03
Team Topic Presentation Development	E	João O., Lucas A., Miguel	Open	17/03

4 Next Meeting

- **Proposed Date:** March 18, 2026
- **Proposed Location:** FEUP I-108

4.1 Next week's main objectives:

- Have a plate selected to buy
- Have the full budget ready
- First experiements with the sound card
- Have a good presentation ready
- Complete the simulator development
- Full system overview scheme